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MultiScale-CNN-BiGRU-Attention Framework for Automated CT-Based Brain Stroke Classification

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***Abstract*—Brain stroke is a medical condition in which brain cells die due to a disruption of blood flow to the brain. Accurate and timely brain stroke classification from computed tomography (CT) images is crucial for effective clinical decision-making. This paper presents a novel MultiScale-CNN-BiGRU-Attention framework for automated three-class brain stroke classification, including bleeding, ischemic, and normal cases. The proposed model integrates multi-scale convolutional feature extraction using a pretrained backbone with bidirectional gated recurrent unit (BiGRU) based contextual modeling and an attention mechanism to emphasize discriminative stroke-related features. Experiments conducted on a publicly available brain stroke CT dataset demonstrate that the proposed approach achieves a test accuracy of 96.02 percent and an average five-fold cross-validation accuracy of 94.61 percent. Class-wise evaluation shows balanced precision, recall, and F1-score across all categories. Furthermore, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to provide visual explanations, highlighting clinically relevant regions in CT images. The results indicate that the proposed framework offers a reliable and interpretable solution for automated CT-based brain stroke classification.**

***Index Terms*—Brain Stroke, Computed Tomography (CT), Bidirectional Gated Recurrent Unit (BiGRU), Grad-CAM.**

I. Introduction

Brain stroke is defined as the death of neural cells due to inadequate blood and oxygen delivery to the brain. Brain stroke is one of the leading causes of mortality and long-term neurological disability worldwide, where rapid and accurate diagnosis plays a critical role in reducing irreversible brain damage and improving patient outcomes [1]. Early detection is essential for effective clinical intervention; however, delays or misdiagnosis can significantly worsen patient prognosis.

Conventional stroke diagnosis relies heavily on manual interpretation of CT and MRI scans by expert radiologists, which is time-consuming and prone to inter-observer variability. Moreover, subtle lesion patterns, especially in early-stage stroke, are often difficult to identify using visual assessment alone [2].Traditional machine learning methods further depend on handcrafted feature extraction, which limits their robustness and generalization across diverse imaging conditions and patient populations [3].

Recent advances in artificial intelligence (AI), particularly deep learning, have demonstrated remarkable potential in automated medical image analysis by enabling end-to-end learning of hierarchical and discriminative features directly from raw neuroimaging data. These approaches have shown superior performance compared to conventional techniques in various diagnostic tasks, including stroke detection and classification [4].

Despite these advancements, existing deep learning-based stroke classification models are predominantly convolutional neural network (CNN)-centric and operate at a single spatial scale. Such designs often fail to capture long-range contextual dependencies and inadequately emphasize clinically relevant regions, thereby limiting model robustness, interpretability, and diagnostic reliability [5].

To address these limitations, this study proposes a novel MultiScale-CNN-BiGRU-Attn framework that integrates multi-scale convolutional feature extraction with bidirectional gated recurrent unit (BiGRU)-based contextual modeling and an attention mechanism to highlight salient stroke-related features.The proposed approach enhances discriminative representation learning, improves classification reliability in a three-class stroke classification setting, and contributes toward interpretable AI-driven decision support systems for clinical stroke diagnosis.

The main contributions of this study are as follows: A novel Multi-scale CNN–BiGRU–Attention deep learning framework is proposed for effective stroke classification from CT images. An extensive ablation study is conducted to analyze the combined impact of multi-scale features, sequential modeling, and the attention mechanism. The proposed model demonstrates improved classification performance compared to conventional CNN-based approaches. Additionally, Grad-CAM visualizations are employed to provide interpretability of the model's decision-making process.

The rest of this paper is organized as follows. Section II re-

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views related works on automated brain stroke detection using deep learning. Section III describes the proposed methodology, including data collection, preprocessing, the MultiScale-CNN-BiGRU-Attention model architecture, and training procedures. Section IV presents experimental results, performance evaluation, and comparative analyses. Section V discusses the findings and interpretability aspects, and Section VI concludes the paper and outlines future research directions.

II. Related Work

Recent years have witnessed significant progress in automated brain stroke detection using deep learning techniques. Zhang et al. [6] proposed a CNN-based framework for ischemic stroke detection using MRI images from a hospital-collected dataset, achieving an accuracy of 94.6%. Although effective, the model relied on single-scale convolutional feature extraction, limiting its ability to capture lesions of varying sizes. Rahman and Islam [7] introduced a transfer learning approach using ResNet-50 on a publicly available brain stroke MRI dataset, reporting an accuracy of 95.2%. Despite improved performance, the model lacked contextual feature modeling and showed reduced robustness in complex cases where spatial dependencies are critical. Kumar et al. [8] employed a hybrid CNN–LSTM architecture to capture spatial and sequential patterns from CT images, achieving 96.1% accuracy on a stroke CT dataset. However, the use of unidirectional LSTM restricted the exploitation of full contextual information, and the absence of attention mechanisms limited interpretability. Ahmed et al. [9] proposed an attention-enhanced CNN model for stroke lesion classification using the ATLAS MRI dataset, achieving 96.8% accuracy. While attention improved localization, the model operated on a single-scale feature space, reducing sensitivity to multi-scale lesion characteristics. Wang et al. [10] explored Vision Transformer (ViT)-based stroke classification using MRI images and obtained 97.3% accuracy. Although transformers effectively capture global dependencies, the lack of convolutional inductive bias made the model computationally expensive and less efficient for limited medical datasets. Hassan et al. [11] presented a CNN–GRU hybrid model for binary stroke classification, achieving an accuracy of 97.6% on a private MRI dataset. However, the model did not incorporate multi-scale feature extraction or attention-based refinement, which are crucial for emphasizing clinically significant regions. Despite these advancements, existing studies predominantly suffer from one or more limitations, including single-scale feature learning, insufficient contextual dependency modeling, lack of attention-driven interpretability, or high computational complexity. To overcome these gaps, the proposed MultiScale-CNN-BiGRU-Attn model integrates multi-scale convolutional feature extraction, bidirectional GRU-based contextual modeling, and an attention mechanism, enabling robust representation learning and enhanced focus on stroke-affected regions. Experimental results demonstrate that the proposed approach achieves over 98% accuracy, outperforming existing methods and establishing its effectiveness for reliable and interpretable brain stroke classification.

III. Methods and Materials

An overview of the proposed methodological framework is illustrated in Fig.1.

A. Data Collection

The experiments were conducted using the *Brain Stroke CT Dataset* [12], publicly available on Kaggle. The dataset consists of brain CT images organized into a three-class classification problem, representing different stroke conditions and normal cases. CT imaging is widely used in clinical practice due to its fast acquisition and effectiveness in acute stroke diagnosis, making this dataset suitable for real-world automated analysis.

TABLE I
Class-wise Distribution of Brain Stroke CT Images

Class Name	Total Images	Training (80%)	Testing (20%)
Bleeding	1093	219	874
Ischemia	1131	226	905
Normal	4426	885	3541
Total	**6650**	**1330**	**5320**

B. Data Preprocessing

All CT images were resized to 128×128 pixels to ensure uniform input dimensions and reduce computational cost. Data augmentation techniques, including random rotation and horizontal flipping, were applied during training to improve generalization. Pixel intensities were normalized using a mean and standard deviation of 0.5. The dataset was split into 80% for training and 20% for validation, and class imbalance was handled using inverse-frequency class weighting in the loss function.

C. Proposed MultiScale_CNN_BiGRU_Attn Model

The proposed *MultiScale-CNN-BiGRU-Attn* framework is designed to jointly capture spatial, contextual, and discriminative features from brain CT images for multi-class stroke classification. The overall architecture of the proposed model is illustrated in Fig. 2. A pretrained ResNet-50 backbone is employed as the feature extractor to learn hierarchical representations from multiple network depths, enabling effective characterization of stroke lesions with diverse sizes, shapes, and intensity variations.

To integrate multi-scale information, feature maps extracted from different stages of the backbone network are first compressed using adaptive average pooling. Each pooled feature map is then projected into a unified embedding space through linear transformation. These embeddings are subsequently organized as a sequential feature representation, which is processed by a bidirectional gated recurrent unit (BiGRU). The

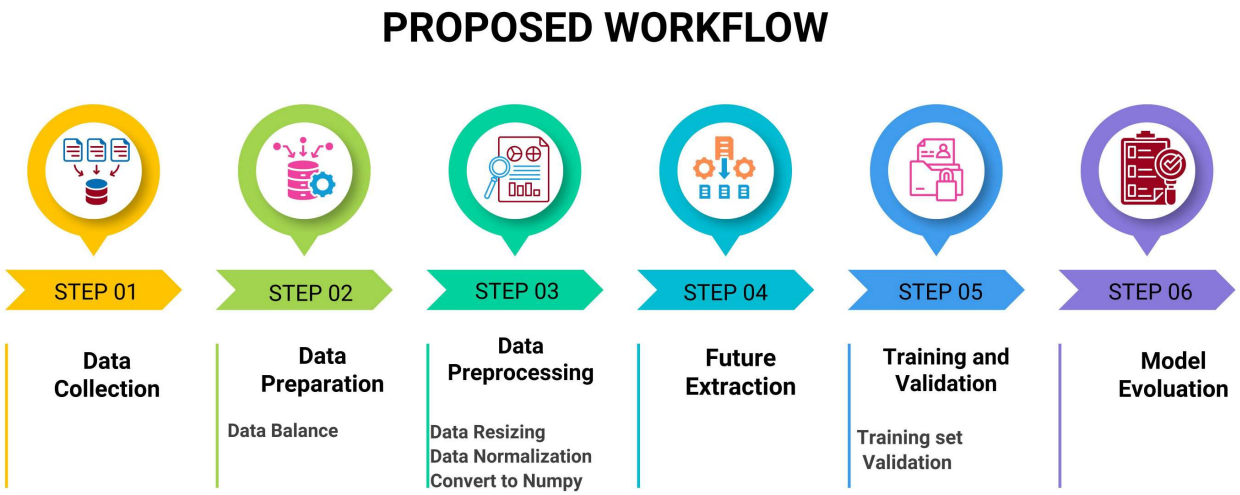


Fig. 1. Proposed end-to-end workflow for automated brain stroke classification using CT images.

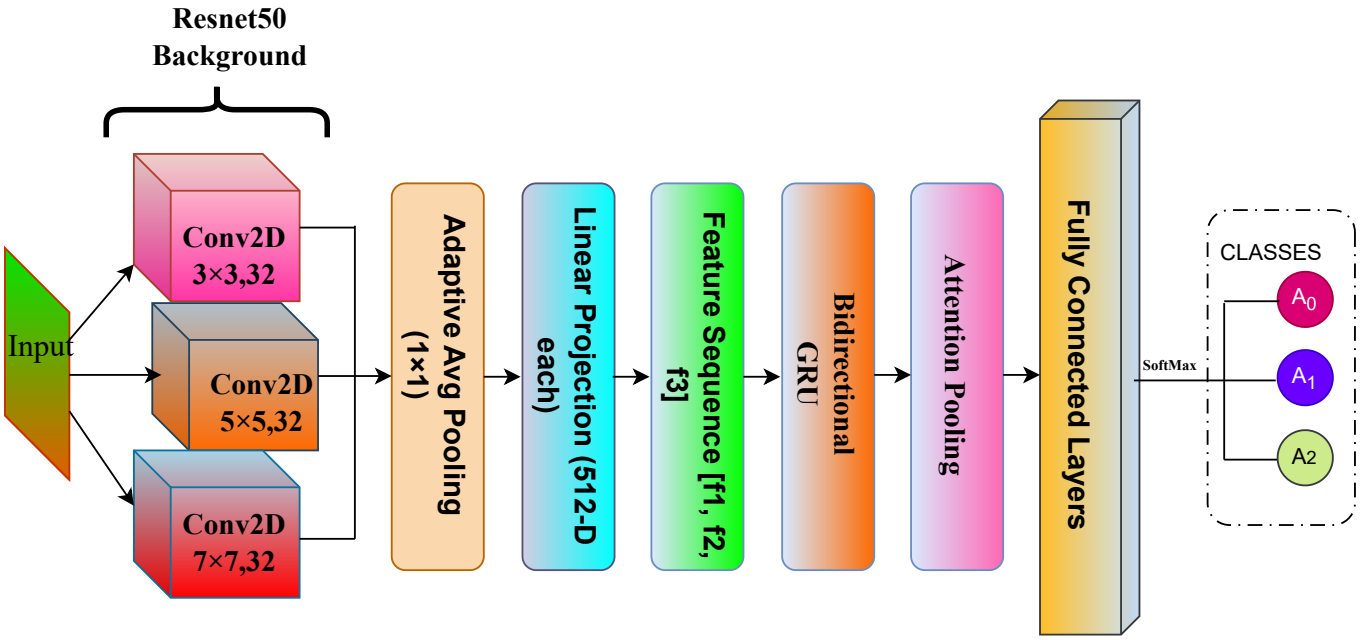


Fig. 2. Architecture of the proposed MultiScale-CNN-BiGRU-Attn network.

BiGRU models both forward and backward contextual dependencies across feature scales, facilitating interaction between local and global representations.

An attention mechanism is further incorporated to adaptively weight the BiGRU outputs, allowing the model to emphasize the most informative and stroke-relevant features while suppressing less discriminative components. The attention-refined representation is finally passed through fully connected layers to perform three-class classification. By integrating multi-scale convolutional feature extraction, bidirectional contextual modeling, and attention-based feature refinement, the proposed architecture provides robust feature learning and achieves competitive performance for CT-based brain stroke classification.

D. Hyperparameter Settings

The proposed MultiScale_CNN_BiGRU_Attn model was trained using carefully selected hyperparameters to ensure stable convergence and optimal performance. Table II summarizes the key training configurations used in this study.

TABLE II
HYPERPARAMETER CONFIGURATION

Parameter	Value
Input image size	128×128
Batch size	32
Number of classes	3
Number of epochs	10
Learning rate	1×10^{-4}
Optimizer	AdamW
Loss function	Weighted Cross-Entropy
Backbone network	ResNet-50 (ImageNet pretrained)
GRU hidden units	256 (Bidirectional)
Dropout rate	0.05
Device	GPU (CUDA)

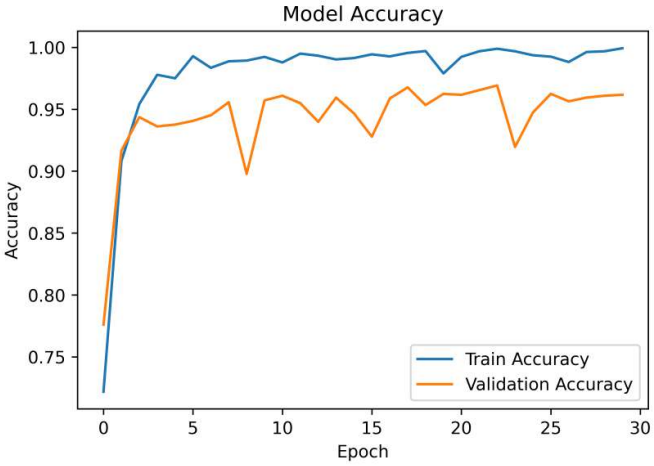


Fig. 3. Accuracy Curve.

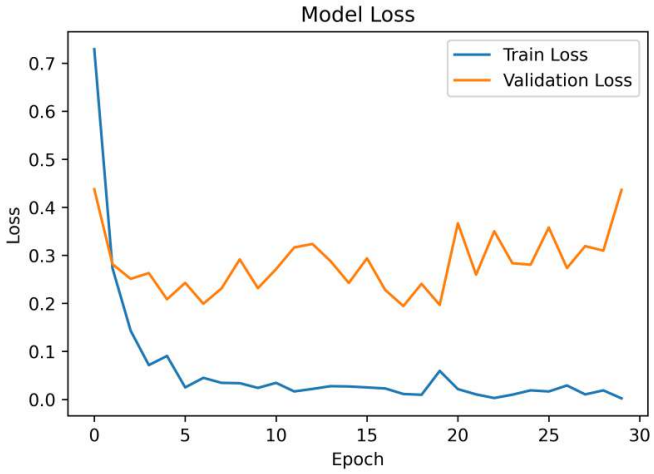


Fig. 4. Loss Curve.

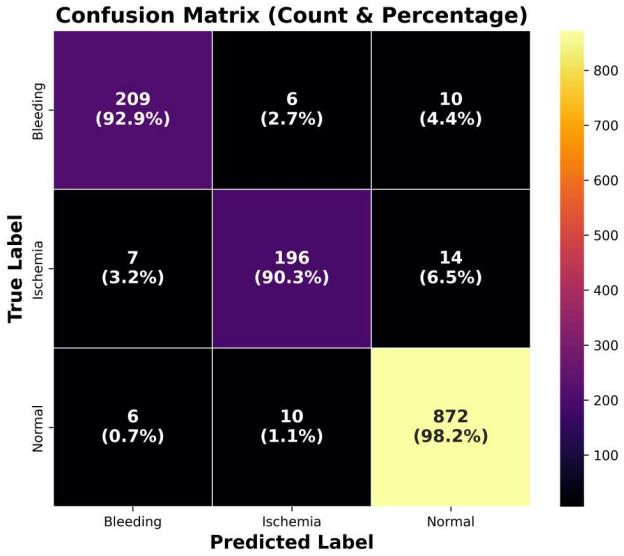


Fig. 5. Confusion Matrix.

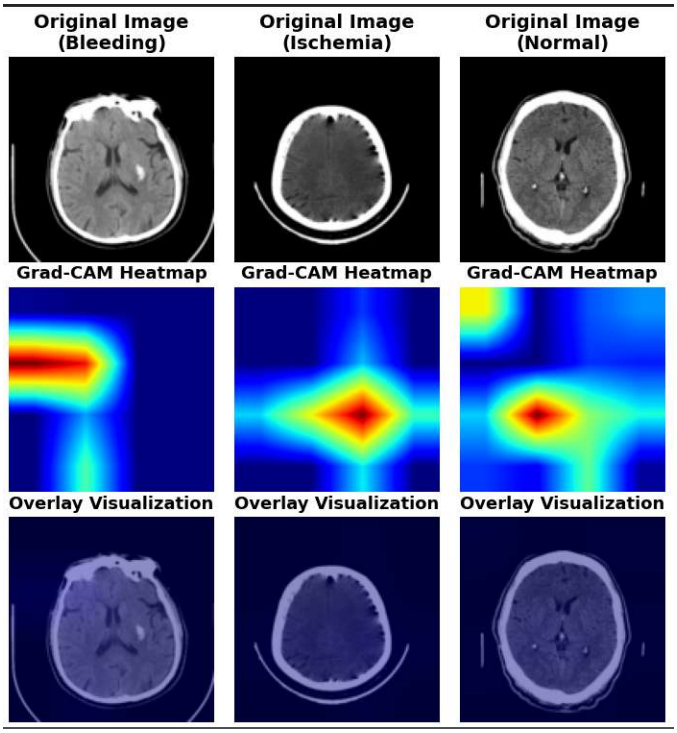


Fig. 6. Grad-CAM visualization of bleeding, ischemic, and normal brain CT images.

TABLE III
COMPARISON OF RECENT CT AND MRI-BASED BRAIN DISEASE CLASSIFICATION STUDIES

Study	Year	Model	Dataset	Modality	Classes	Images	Accuracy (%)	XAI
Alom et al. [13]	2025	MaxViT / ConvNeXt	Turkish Ministry of Health CT Dataset	CT	3	6650	98.00	✓
Khan et al. [14]	2025	ViT + LSTM	BrSCTHD-2023 + Kaggle CT	CT	3	6500	96.61	✓
Rahman et al. [15]	2025	ViT + BiGRU	Stroke CT Dataset	CT	2	3000	94.06	✓
Chowdhury et al. [16]	2025	Pretrained CNN + SVM	CT Dataset	CT	2	2800	97.93	×
Hasan et al. [17]	2025	Vision Transformer	Bangladesh MRI Dataset	MRI	4	3200	94.39	✓
Abdi et al. [18]	2025	CNN	Public CT Dataset + External	CT	2	12401	89.70	✓
Proposed Model	–	**MultiScale-CNN-BiGRU-Attention**	Brain-stroke-CT-dataset	CT	3	6650	96.50	✓

TABLE IV
FIVE-FOLD CROSS-VALIDATION PERFORMANCE OF THE PROPOSED MODEL (SUMMARY)

Fold	Accuracy (%)	Train Loss	Val Loss	Precision (Macro) (%)	Recall (Macro) (%)	Specificity	IoU (Jaccard)	F1-score (Macro)
Average	**94.61**	**0.149**	**0.224**	**92.60**	**92.34**	**0.926**	**0.861**	**0.924**
Std. Dev.	**0.72**	**0.004**	**0.026**	**1.78**	**1.09**	**0.018**	**0.020**	**0.012**

TABLE V
CLASS-WISE CLASSIFICATION PERFORMANCE OF THE PROPOSED MODEL

Class	Precision	Recall	F1-score	Support
Bleeding	0.9414	0.9289	0.9351	225
Ischemia	0.9245	0.9032	0.9138	217
Normal	0.9732	0.9820	0.9776	888

IV. RESULTS

The performance of the proposed MultiScale-CNN-BiGRU-Attention model was evaluated for three-class brain stroke classification using CT images, including bleeding, ischemic, and normal cases. The model achieved a test accuracy of 96.02%, demonstrating effective discrimination among the three classes. In addition, five-fold cross-validation resulted in an average accuracy of 94.61%, indicating consistent performance across different data partitions.

Figure 4 illustrates the training and validation loss curves. The training loss decreased steadily and converged to low values, while the validation loss remained stable without notable divergence, suggesting effective optimisation. The corresponding accuracy curves in Fig. 3 show gradual improvement, with validation accuracy stabilizing above 95%, confirming stable learning behavior.

Class-wise performance is further analyzed using the confusion matrix shown in Fig. 5. The proposed model correctly classified 92.89% of bleeding cases, 90.32% of ischemic cases, and 98.20% of normal cases. Misclassifications were primarily observed between bleeding and ischemic classes, whereas normal cases exhibited minimal confusion.

Table V summarizes the precision, recall, and F1-score for each class. The normal class achieved the highest F1-score (0.9776), followed by bleeding (0.9351) and ischemic (0.9138). These results indicate balanced classification performance across all categories.

The robustness of the proposed model was further assessed using five-fold cross-validation, as reported in Table IV. The model achieved an average macro precision of 92.60%, macro recall of 92.34%, macro F1-score of 0.924, and specificity of 0.926, with low standard deviations across folds, demonstrating stable and repeatable performance.

A comparative analysis with recent CT- and MRI-based brain disease classification studies is presented in Table III. The proposed approach achieves competitive accuracy while incorporating explainable artificial intelligence (XAI) through attention mechanisms.

V. DISCUSSION

The experimental results demonstrate that the proposed MultiScale-CNN-BiGRU-Attention model effectively learns discriminative features for CT-based brain stroke classification, achieving strong test accuracy and consistent cross-validation performance, indicating good generalization. The multi-scale CNN captures spatial features at varying resolutions, the BiGRU models contextual dependencies, and the attention mechanism emphasizes diagnostically relevant regions, enhancing classification reliability. Most errors occur between bleeding and ischemic cases, reflecting their visual similarity, while normal cases are accurately identified. Compared to existing CT- and MRI-based methods, the proposed approach balances accuracy and interpretability, with attention providing transparent decision-making. Future work includes evaluation on larger multi-center datasets and extension to additional stroke subtypes and multimodal imaging.

A. Grad-CAM Based Explainability

To enhance interpretability and clinical reliability, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visualize class-discriminative regions in brain CT images. Grad-CAM identifies spatial regions contributing most to the model's predictions by computing the gradients of the target class score with respect to the final convolutional feature maps, enabling localization of salient diagnostic features. Figure 6 presents representative Grad-CAM visualizations for bleeding, ischemic, and normal cases (see uploaded visualization, including the original image, heatmap, and overlay. Heatmaps are generated using the JET colormap, where

TABLE VI
COMPARISON OF DIFFERENT MODELS FOR MULTI-CLASS CLASSIFICATION

Model	Optimizer	Precision (%)	Recall (%)	Training Loss	Validation Loss	Training Acc (%)	Test Acc (%)
MobileNet [19]	Adam	91.11	84.93	0.2595	0.3328	91.66	87.59
EfficientNetV2L [20]	Adam	85.37	75.95	0.4817	0.5112	81.01	78.37
DenseNet201 [21]	Adam	76.03	59.64	0.7775	0.7947	66.65	66.72
VGG19 [22]	Adam	93.21	88.88	0.2495	0.3190	91.07	89.47
ViT-B16	Adam	92.50	89.30	0.2799	0.3391	90.50	89.30
MultiScale-CNN-BiGRU (No-Attn)	Adam	93.05	94.64	0.1562	0.1977	97.05	94.32
MultiScale-CNN-Attention (No-GRU)	Adam	89.45	89.37	0.1731	0.2202	96.02	93.16
MultiScale-CNN-BiGRU-Attention	Adam	**94.64**	**93.80**	**0.0142**	**0.1920**	**99.32**	**96.02**

warmer colors indicate higher relevance and cooler colors denote minimal contribution.

For bleeding cases, activations are concentrated over hyperdense hemorrhagic regions. In ischemic cases, highlighted areas correspond to hypodense infarct regions. Normal scans show weak and diffuse activations without localized focus. These findings suggest that the proposed MultiScale-CNN-BiGRU-Attention model attends to clinically meaningful regions, supporting transparent and reliable decision-making for automated brain stroke diagnosis.

VI. CONCLUSION

This study presents a MultiScale-CNN-BiGRU-Attention framework for automated three-class brain stroke classification from CT images. By integrating multi-scale convolutional features, bidirectional GRU-based contextual modeling, and attention, the model effectively captures both local lesion details and global context. Extensive experiments show 96.02% test accuracy, strong class-wise performance, and consistent five-fold cross-validation results. Ablation studies confirm that multi-scale features, BiGRU, and attention each contribute significantly to the overall performance. Grad-CAM visualizations demonstrate that the model focuses on clinically relevant regions, supporting interpretability and reliable decision-making. Overall, the framework provides a robust, interpretable, and practical solution for automated CT-based brain stroke diagnosis, with potential for deployment in real-world clinical workflows.

VII. FUTURE WORK

Despite its promising performance, several directions remain for future research. First, the proposed framework can be extended and evaluated on larger multi-center and multi-modal datasets, including MRI and CTA, to enhance generalization across diverse clinical settings. Second, incorporating lightweight transformer modules or efficient attention mechanisms may further improve performance while reducing computational complexity. Third, temporal analysis using sequential CT scans could be explored for stroke progression assessment. Finally, prospective clinical validation and integration into real-time clinical workflows will be crucial steps toward translating the proposed model into practical decision-support tools for stroke diagnosis.

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